

Requirements for Final Project of C++ Programming

In principle, each student shall complete the project independently. Effective collaboration is also encouraged (collaboration is more challenging than independent work). Each group shall consist of 1-2 members. Group members must specify their respective responsibilities in the project proposal report, and all members must participate in individual final Demo.

The initial schedule as following:

May 7th: submit your choice of the final

May 27th: Tutorial III: A complete chess game

June 8th: Submit the project

June 10th: Demo session

Knowledge points covered in your implementation:

Mandatory: (total score will not exceed 60 if requirements are not met)

1. Classes, File I/O
2. Multi-file structure: .h and .cpp files corresponding to class structures (passing criteria; total score will not exceed 60 if requirements are not met)

Optional: (The grades will based on these issues.)

1. Inheritance, Polymorphism, Arrays, Linked Lists, Templates.....
2. Core implementation shall be in C++; library calls or support from other languages are allowed if necessary.

Project Topic

Free choice of topic.

It could be a chess game. Comparing with the chess game introduced in the class, you may need to implement in a even better one in terms of the functionality and better design.

You can also refer to the Raylib examples.

- Raylib Examples: <https://www.raylib.com/examples.html>

Submission of Project Proposal (May 7th)

Fill in the information about your final project.

https://lbwcs29n19.feishu.cn/sheets/Dq0YsLsXU9eghtzvR2cHooynd?from=fro_m_copylink

Final Submission of Complete Project (June 8th)

Submit the following documents:

- Project Proposal and Requirements Analysis Report
- Design Specification
- User Manual
- Program Source Code (with comments)
- Test Cases (mandatory, no score)
- Summary Report (mandatory, no score)

Final Defense (June 10)

- Presentation of documents and source code
- Demonstration of program functions
- Answering questions from instructors

Grading Items (Total 100 points)

1. Design Specification (30 points)
3. User Manual (20 points)
4. Program Code Implement (50 points)
5. Test Cases and Screen Recording of the demo (mandatory; 1-5 points deduction if missing or unreasonable)
6. Summary Report (mandatory; 1-5 points deduction if missing or unreasonable)
7. Code Style and Rationality (random inspection; 1-20 points deduction for unreasonable implementation)

Detailed Requirements

1 Design Specification (30 points)

Class Design

- Class Diagram: Clearly show class members and relationships between classes.
- Data Member Description: Explain what data is stored, its meaning and purpose.
- Function Member Description: Specify function prototypes, detailed descriptions, and main functions.

2 Program Implementation (50 points)

- Main Technical Difficulties and Implementation Solutions/Algorithm Design Software Structure (additional deduction if structure is unclear due to poor specification)
- Rational Module Division: Each module performs a single task; reusable functions separated; classes correspond to real-world objects or design requirements.
- Low Coupling: Minimal interdependence; limited use of global objects/variables;

lvalues, non-const pointers, and non-const references used only when necessary.

- Good Encapsulation: Public members and non-const functions used only when necessary.

- Rational Class Relationships: Appropriate use of composition, inheritance, and polymorphism.

- Proper File I/O Usage: Projects with no file I/O must explain why, and demonstrate file I/O with unrelated data.

3. User Manual (20 points)

System operation instructions enabling users to use the program correctly.

Clearly specify restrictions on operations and input data. Insufficient instructions will lead to deductions; unreasonable tests based on unstated restrictions will cause further deductions in defense.

4. Test Cases and Software Usage Screen Recording (mandatory, no score; 1-5 points deduction if missing/unreasonable)

Thoroughly test the program with diverse data and workflows, covering all functions and branches. Record test cases and results. For game software, describe test content and include screenshots. Submit complete screen recording as video file or cloud link. Prepare demonstration data for defense.

5. Summary Report (mandatory, no score; 1-5 points deduction if missing/unreasonable)

Describe project experience, challenges, and achievements.

Detail AI usage: tasks completed with AI, iterative improvements, and problems solved.

6. Code Style and Rationality (random inspection; 1-15 points deduction)

Deductions for poor practices: global variable abuse, weird naming, overly long functions, unclear logic, redundant lines, oversized files.

7. AI Usage Description

Report AI usage in framework construction and implementation, understanding level, and issues encountered. 1-15 points deduction for over-reliance without modification or optimization.